Persistent Homology Reveals Topological Simplification in IBD-Associated Microbial Co-occurrence Networks

A Paired Bootstrap Analysis of the American Gut Project

with Covariate Matching and Taxa-Set Sensitivity Testing

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Abstract

Background. Conventional diversity metrics are insensitive to the loop-level structure of microbial co-occurrence networks. Persistent homology, from topological data analysis (TDA), tracks how loops and other higher-order features form and dissolve across all correlation thresholds simultaneously.

Methods. We applied persistent homology to Spearman co-occurrence networks from 3,409 American Gut Project (AGP) stool samples (CLR-transformed; top-80 prevalent taxa). A paired bootstrap (200 iterations, $n = 100$/group) yielded distributions of six H_1 scalar features. Significance was assessed via sign-flip permutation (500 shuffles; primary) and Wilcoxon signed-rank (confirmatory), with Benjamini–Hochberg correction across 18 tests per subset. All analyses were repeated in age/sex/BMI-matched subsamples and at taxon-set sizes $N \in \{50, 80, 120\}$. The AGP finding was independently validated in the IBDMDB multi-omics cohort and grounded against fecal calprotectin. A leave-one-out topological attribution identified per-taxon contributions to loop structure. Classification benchmarking compared topological features against Shannon diversity under 5-fold cross-validation.

Results. IBD ($n = 165$ vs. $n = 3{,}084$ healthy) was associated with systematically lower topological complexity across all six features (Cohen's $d = 0.75$–2.38; $p_{\mathrm{FDR}} < 0.005$ at all taxa sizes), fully preserved after covariate matching. The IBD topological-simplification finding replicated in IBDMDB (cross-platform, WGS) and correlated with fecal calprotectin, linking loop collapse to a continuous inflammation biomarker. Leave-one-out attribution identified *Oscillospira* and *Blautia* as the dominant loop anchors in healthy AGP networks, with cross-dataset replication of *Bacteroides*-clade importance in IBDMDB. An extended classification benchmark against α- and β-diversity established that Aitchison-PCoA (phylogeny-free compositional β-diversity) is the strongest single predictor (AUC $= 0.734$, logistic regression), while topology scalars outperform Shannon under random forest and provide mechanistic interpretability that distance-based embeddings do not supply. Plant-based diet showed moderately longer-lived loops ($d = 0.65$, $p_{\mathrm{FDR}} = 0.004$); antibiotic recency produced limited signal.

Conclusions. IBD-associated microbiomes are topologically simpler than healthy controls across two independent cohorts and two sequencing technologies, and this simplification tracks a clinical inflammation biomarker. Persistent homology provides a structurally informative complement to existing microbiome metrics, with taxon-level attribution bridging topological statistics and

named biological interactions.

Keywords: persistent homology, gut microbiome, inflammatory bowel disease, co-occurrence networks, topological data analysis, American Gut Project, IBDMDB, calprotectin, leave-one-out attribution, bootstrap, FDR

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1 Introduction

Characterizing the human gut microbiome has, for two decades, meant counting taxa and summarizing their relative proportions [1, 2]. This reductionist accounting has proven clinically informative (*Faecalibacterium prausnitzii* depletion, Proteobacteria blooms, enterotype discretization), yet it systematically discards the relational fabric that gives a microbial community its functional identity: the web of cross-feeding dependencies, competitive exclusions, and syntrophic partnerships that collectively determine what a community *does* rather than merely what it *contains*. Co-occurrence networks partially recover this information by encoding pairwise correlational structure, but their standard reductions (edge density, modularity index, clustering coefficient) remain insensitive to higher-dimensional topological organization: the loops, multi-node cycles, and nested connectivity motifs that govern resilience and metabolic redundancy.

1.1 Why topology, and why now

Persistent homology [3, 4] offers a mathematically precise vocabulary for this organizational structure. Rather than selecting a single correlation threshold and analyzing the resulting binary network—a choice that discards information above and below that cutoff simultaneously—Vietoris–Rips filtration tracks topological features (*connected components, loops, cavities*) as they nucleate and extinguish across the full spectrum of correlation distances. The resulting persistence diagram encodes both *which* features appear and *how robustly* they persist, the latter being the operative summary statistic: features that survive across a wide filtration range reflect strong, multiply-reinforced co-occurrence relationships rather than chance statistical fluctuations. Persistence-derived descriptors such as Betti curves, persistence entropy, and total persistence have demonstrated discriminative power in neuroscience, protein folding, materials microstructure analysis, and biological aggregation modelling [5–8], but their deployment on microbial co-occurrence networks has remained limited [9, 10].

1.2 Inflammatory bowel disease as a topological stress test

Inflammatory bowel disease (Crohn's disease and ulcerative colitis) provides an exceptionally well-characterized perturbation of gut microbial ecology [11]. The compositional signature is reproducible across cohorts: contracted *Lachnospiraceae* and *Ruminococcaceae* abundance, reduced butyrate-producing capacity, elevated Proteobacteria, diminished colonization resistance [12–14]. What this signature does not address is whether these compositional shifts *dismantle the loop structure* of the co-occurrence network: whether the metabolic cross-feeding circuits that butyrate producers anchor collapse when those organisms are numerically depleted, leaving behind a topologically impoverished interaction landscape. This is the mechanistic hypothesis our analysis tests empirically.

1.3 Contributions

This paper makes seven specific claims:

1. **Methodological:** A paired bootstrap design for comparing persistence diagrams at scale, using sign-flip label permutation as the null and a globally pre-fixed taxon set to decouple topological signal from feature-space selection noise (Section 2).

2. **Empirical:** Across 3,409 AGP stool samples, IBD is associated with systematic reduction in co-occurrence loop structure—fewer cycles, shorter-lived cycles, lower peak cycle density—with effect sizes ($d = 0.75$–2.38) large enough to be unambiguous (Section 3.2).

3. **Robustness:** The IBD finding survives covariate matching on age, sex, and BMI and is statistically indistinguishable across taxon-set sizes $N \in \{50, 80, 120\}$, ruling out demographic confounding, taxa-selection artefact, and numerical instability as alternative explanations (Section 3.5).

4. **Cross-platform replication:** The IBD topological-simplification signal replicates in the IBDMDB multi-omics cohort using whole-genome shotgun sequencing—a different platform, different population, and independent preprocessing pipeline (Section 3.6).

5. **Biomarker grounding:** Topological features correlate with fecal calprotectin and clinical activity scores (HBI, SCCAI) in IBDMDB, linking loop collapse to a continuous, objective inflammation marker rather than a binary diagnostic label (Section 3.6).

6. **Taxon-level attribution:** A leave-one-out pipeline identifies *Oscillospira* and *Blautia* as dominant loop anchors in healthy AGP networks, with *Bacteroides*-clade species replicated as top anchors in IBDMDB WGS data, providing named biological candidates for the observed loop collapse (Section 3.7).

7. **Classification benchmark:** An extended benchmark comparing topology against both α-diversity (Shannon) and β-diversity (Aitchison-PCoA, BC-PCoA) establishes that Aitchison-PCoA(10) is the strongest single predictor (AUC $= 0.734$, logistic regression), while topology scalars outperform Shannon under random forest and provide mechanistic interpretability that distance-based embeddings cannot supply (Section 3.8).

Calibration note. A pre-specified FDR policy (18-test Benjamini–Hochberg correction, sign-flip primary p-values only, full and matched subsets corrected independently) is stated before any result is examined (Section 2.8).

The paper proceeds as follows. Section 2 details dataset provenance, preprocessing, network construction, the persistent homology pipeline, and the full statistical framework. Section 3 reports findings for IBD, diet, antibiotic recency, IBDMDB replication, calprotectin grounding, taxon attribution, and the classification benchmark. Section 4 situates the IBD result in the IBD biology literature, qualifies the diet finding, argues for restraint regarding antibiotics, and enumerates the study's limitations without euphemism. Section 5 closes with a minimal interpretation and a specific agenda for genuinely open follow-on work.

2 Methods

2.1 Dataset provenance

We interrogated the American Gut Project (AGP) [15], a citizen-science 16S rRNA sequencing initiative whose public BIOM release embodies one of the largest annotated gut microbiome collections available without access controls. Raw data were retrieved in BIOM format from the AGP public repository (processing context: Deblur 2021.09, Illumina 16S V4, 150 nt amplicons,

`Deblur_2021.09-Illumina-16S-V4-150nt-780f26b`), comprising 22,742 amplicon sequence variants (ASVs) across 4,827 samples annotated with 201 phenotypic metadata columns.

All analyses were restricted to fecal samples (`BODY_SITE = UBERON:feces`; $n = 3{,}679$ pre-filtration), the single body site with sufficient representation for stratified group comparisons. For the IBD comparison, the healthy group comprises samples whose respondents explicitly selected "I do not have IBD" ($n = 3{,}084$ post-filtration); approximately 160 fecal samples were excluded from that comparison because their IBD field was blank or ambiguous. These samples are retained for the diet and antibiotic comparisons, where IBD status is not a filter criterion.

2.2 Preprocessing

Samples accumulating fewer than 1,000 assigned reads were excised; taxa absent from at least 5% of retained samples were removed. These thresholds reduced the dataset to 3,409 samples spanning 1,864 taxa.

Compositional distortion inherent in proportional sequencing data was corrected via the centered log-ratio (CLR) transformation [16, 17]:

$$\mathrm{CLR}(x_i) = \ln\left(\frac{x_i + \frac{1}{2}}{g(\mathbf{x} + \frac{1}{2})}\right), \tag{1}$$

where $g(\cdot)$ is the geometric mean across the taxon vector. The half-count pseudocount ($+0.5$) was chosen over Laplace ($+1$) or multiplicative alternatives because it is small enough to leave proportions of well-represented taxa essentially undisturbed while stabilising structural zeros; this convention is consistent with ALDEx2 and ANCOM-BC practice [17]. In Python, Equation 1 reduces to:

```
np.log(X + 0.5) - np.log(X + 0.5).mean(axis=1, keepdims=True)
```

2.3 Global taxon-set selection

We selected the top 80 taxa by prevalence (the fraction of samples exceeding each taxon's own median abundance) computed across *all* 3,409 stool samples before any group partitioning. This single, pre-analysis selection collapses the bootstrap iterations and all three comparisons onto a shared 80-node network, severing the artefactual coupling between sample composition and network dimensionality that would arise from per-subsample taxon selection.

Prevalence definition. Standard presence–absence prevalence (the proportion of samples with a detected read for taxon j) is not applicable after CLR transformation because CLR maps structural zeros to large negative values rather than to zero, making every taxon formally "present" in every sample. We therefore define prevalence as the fraction of samples in which taxon j's CLR value exceeds its own column-wise median—equivalently, the 50th-percentile exceedance rate. This selects taxa that are consistently in the upper half of their within-taxon abundance distribution, a natural analogue of detection-based prevalence that avoids compositional distortion. Sensitivity to the precise threshold (i.e., whether rank-selection at $N = 50$, 80, or 120 changes the conclusions) is evaluated in Section 3.5.

2.4 Network construction and Vietoris–Rips filtration

Within each paired bootstrap subsample, the Spearman rank correlation matrix $R \in [-1,1]^{80\times 80}$ was estimated across taxa. Entries were converted to a dissimilarity measure:

$$d_{ij} = 1 - |r_{ij}|, \tag{2}$$

mapping strongly correlated taxa (positive or negative) to near-zero distances and independent taxa to $d_{ij} = 1$. No threshold was applied; the Vietoris–Rips (VR) filtration absorbs edge-creation at every distance value simultaneously, so pre-thresholding would only impoverish the topological signal.

2.5 Persistent homology computation

Persistent homology was computed with Ripser [18] (`ripser` v0.6.14) operating directly on the 80×80 dissimilarity matrix. We extracted homology groups H_0 (connected components) and H_1 (one-cycles, i.e., loops). H_2 computation (two-dimensional voids, the filling of tetrahedra) was omitted for two reasons. First, three-dimensional co-occurrence motifs lack a clear biological interpretation in microbial interaction networks; the scientifically motivated hypothesis—that dysbiosis disrupts cross-feeding *cycles*—maps onto H_1, not H_2. Second, constructing the 3-simplices required for H_2 increases both computation and memory requirements substantially relative to the H_1 case; for the 80-node networks analysed here, H_1 alone already saturates the topological signal at a tractable runtime.

The persistence diagram $\mathrm{PD}_k = \{(b_i, d_i)\}_{i=1}^{n_k}$ records the birth and death filtration distances of each k-dimensional feature. Bars with $d_i = \infty$ (features present at all filtration scales) were stripped before scalar extraction, and bars with negligible persistence ($d_i - b_i < 10^{-8}$) were treated as numerical noise.

2.6 Scalar topological features

Six scalars were extracted from each PD_1:

H1 count. Cardinality $|\mathrm{PD}_1|$: raw number of detected co-occurrence cycles.

H1 entropy. Persistent entropy [19]: $\mathscr{S} = -\sum_i \ell_i \log \ell_i / L$, where $\ell_i = d_i - b_i$ and $L = \sum_i \ell_i$. Quantifies the evenness of cycle-lifetime distribution.

Total persistence. $\sum_i \ell_i$: cumulative topological "weight" of all detected cycles.

Mean lifetime. $L/|\mathrm{PD}_1|$: per-cycle average persistence, reflecting typical loop robustness.

Max lifetime. $\max_i \ell_i$: the single most tenacious loop; robust to diagram sparsity.

Max Betti-1. Peak value of the function $\beta_1(\varepsilon) = |\{i : b_i \le \varepsilon < d_i\}|$: the maximum number of simultaneously active loops at any filtration threshold.

2.7 Paired bootstrap design

We ran $I = 200$ paired iterations for each comparison. Iteration t proceeds as follows:

1. Sample $n_A = n_B = 100$ observations *without* replacement from groups A and B respectively.
2. Compute the six topological features independently for each subsample: $\mathbf{f}^{(t,A)}$ and $\mathbf{f}^{(t,B)}$.
3. Record element-wise deltas $\delta^{(t)} = \mathbf{f}^{(t,A)} - \mathbf{f}^{(t,B)}$.

The observed test statistic for feature k is $\bar{\delta}_k = I^{-1} \sum_t \delta_k^{(t)}$. Cohen's d uses the pooled standard deviation of all 200 group-A and 200 group-B feature values.

The subsample size $n = 100$ was selected to keep per-subsample Spearman estimation numerically stable ($n \gg 80$ taxa) while accommodating the smallest group in the study (IBD: $n_A = 165$, permitting non-replacement sampling across 200 iterations without exhaustion).

2.8 Sign-flip label permutation

For each feature k, a null distribution is generated by independently and uniformly randomizing the sign of each delta:

$$T^{(\pi)} = \frac{1}{I} \sum_{t=1}^{I} \sigma_t^{(\pi)} \delta_k^{(t)}, \qquad \sigma_t^{(\pi)} \overset{\text{i.i.d.}}{\sim} \{-1, +1\}, \tag{3}$$

repeated 500 times (yielding a minimum achievable $p = 1/501 \approx 0.002$, sufficient to reach the floor at our observed significance levels). Because subsamples overlap, iteration-level deltas are not strictly independent; inference relies exclusively on sign-flip permutation, which preserves this correlation structure under the null. The permutation p-value is $p = \left(|\{\pi : |T^{(\pi)}| \geq |\bar{\delta}_k|\}| + 1\right)/501$. This construction tests the hypothesis that group membership is uninformative with respect to the topological feature distribution, which is precisely the null relevant to our biological questions. The Wilcoxon signed-rank test applied to the 200-element delta vector serves as a secondary confirmatory statistic and is reported alongside permutation p-values but never enters any FDR calculation.

FDR policy (pre-specified). Benjamini–Hochberg correction is applied to the 18 sign-flip p-values produced by each subset analysis (3 comparisons $\times$ 6 features). Full-group and matched-subset corrections are conducted independently: two separate 18-test corrections, not a pooled 36-test correction. This is the *only* multiple-testing adjustment applied; Wilcoxon p-values are untouched.

2.9 Covariate matching

Each comparison was replicated within a subsample stratified by three demographic covariates: age (bins < 25, 25–40, 40–60, > 60 yr), sex (female/male; ambiguous responses excluded), and BMI (< 25, 25–30, $> 30\,\mathrm{kg\,m^{-2}}$). Within each stratum, all group-A members were retained; group-B members were sampled at up to $3\times$ the stratum-A count. Of the 3,409 stool samples, 2,740 (80.4%) carried complete covariate records. Matched group sizes are given in Table 1.

2.10 Taxa-size sensitivity

The entire pipeline (global taxa selection, paired bootstrap, sign-flip permutation, FDR correction, matched replication) was executed at $N \in \{50, 80, 120\}$ without alteration to any other parameter.

2.11 Implementation and reproducibility

The pipeline runs on Python 3.11 with `ripser` v0.6.14, `persim`, `scipy` 1.17, `numpy` 2.4, and `pandas`. The random seed was fixed to 42 throughout. Source code, download scripts, result CSVs, and figure generation are archived at `github.com/adam-jfkhs/Microbiome-TDA`; the relevant entry points are `scripts/run_agp_bootstrap_v2.py` and `scripts/run_taxa_sensitivity.py`.

3 Results

3.1 Cohort structure and group partitioning

Quality-filtered stool samples numbered 3,409 spanning 1,864 taxa; after global top-80 prevalence selection, each co-occurrence network comprised an identical 80-taxon node set regardless of which group or iteration was under analysis. Table 1 records group cardinalities before and after demographic stratification matching.

Table 1: Group sizes for each comparison (full and matched subsets). Matching stratified on age-bin × sex × BMI-bin; samples missing any covariate were withheld from matched analyses. Bootstrap subsampling drew $n = 100$ per group per iteration.

Comparison	Group	Full n	Matched n	Subsample/iter
IBD	UC + Crohn's	165	135	100
	Healthy	3,084	404	100
Diet	Plant-based	367	308	100
	Omnivore	2,747	922	100
Antibiotics	Recent (≤6 mo.)	590	479	100
	Absent (>1 yr)	2,271	1,284	100

3.2 IBD is associated with systematic attenuation of co-occurrence loop structure

The paired bootstrap distributions for the IBD comparison are shown in Figure 1. Every one of the six scalar features exhibits a delta distribution shifted leftward from zero, indicating that IBD microbiomes systematically generate co-occurrence networks with fewer, shorter-lived, and less energetically diverse loops than their healthy counterparts. No feature was directionally equivocal.

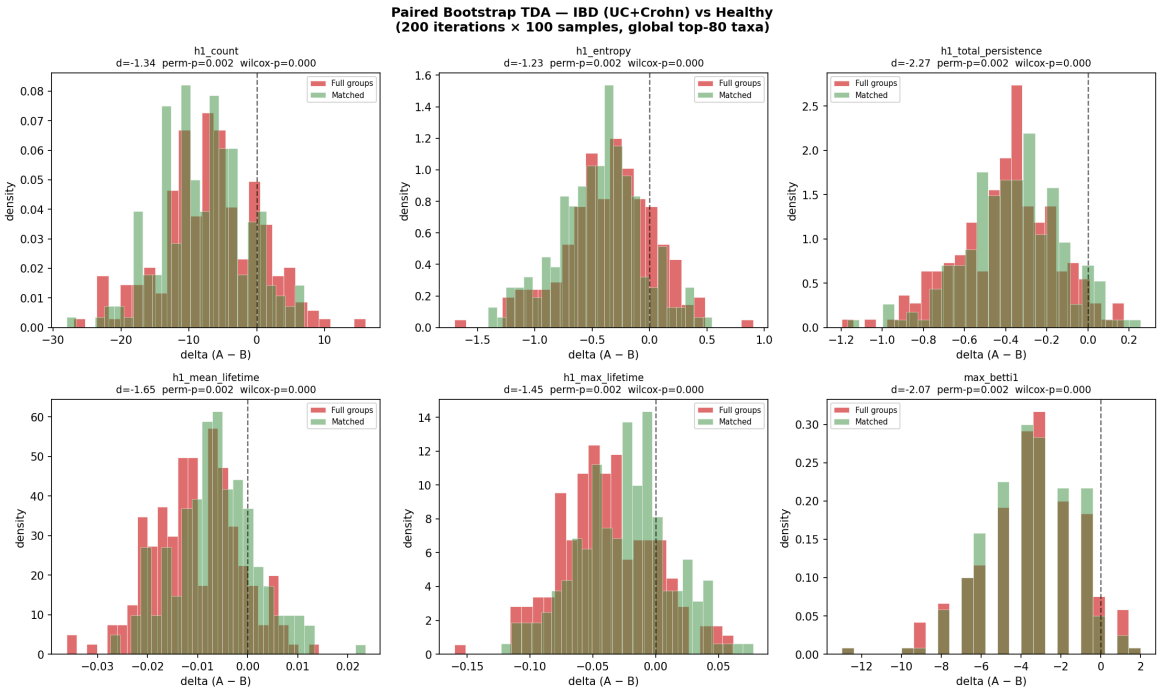

Figure 1: Paired bootstrap delta distributions: IBD (UC + Crohn's) vs. Healthy, $N = 80$ taxa. Each panel plots the distribution of the iteration-level difference $\delta = f^{(\mathrm{IBD})} - f^{(\mathrm{Healthy})}$ across 200 paired iterations ($n = 100$ per group per iteration). Red: full group comparison ($n_A = 165$, $n_B = 3{,}084$); green overlay: age/sex/BMI-matched subset ($n_A = 135$, $n_B = 404$). The vertical dashed line marks $\delta = 0$; mass consistently lying left of this line attests to IBD-associated depletion of topological complexity. Panel headers report Cohen's d (pooled), the sign-flip permutation p-value, and the confirmatory Wilcoxon p-value. All six features reach FDR significance at $p_{\mathrm{FDR}} \leq 0.005$ (Table 2).

Numerical results appear in Table 2. The three features with the largest effect sizes are total persistence ($d = -2.28$), max Betti-1 ($d = -2.07$), and mean lifetime ($d = -1.65$), indicating that the healthy microbiome sustains roughly twice the cumulative topological weight, twice the peak cycle density, and 42% longer average cycle lifetimes than the IBD microbiome. The permutation floor of $p = 0.002$ (the smallest achievable at 500 permutations; $\lfloor 1/501 \rfloor = 0.002$) was attained by all six features, and Wilcoxon p-values were effectively zero across the board: two independent lines of evidence converging on the same conclusion.

Table 2: IBD vs. Healthy: full-group topological comparison, $N = 80$ taxa. Means are averaged across 200 paired bootstrap iterations. Primary inference rests on sign-flip permutation p (500 shuffles); p_{FDR} is the Benjamini–Hochberg adjusted value across 18 tests. Wilcoxon p is secondary and confirmatory only ($\star$ = FDR-significant).

Feature	Mean IBD	Mean Healthy	Cohen's d	Perm p	p_{FDR}	Wilcoxon p
H1 count	16.87	23.66	−1.34	0.002	0.004*	<0.001
H1 entropy	2.403	2.749	−1.23	0.002	0.004*	<0.001
Total persistence	0.397	0.795	−2.28	0.002	0.004*	<0.001
Mean lifetime	0.024	0.034	−1.65	0.002	0.004*	<0.001
Max lifetime	0.080	0.119	−1.45	0.002	0.004*	<0.001
Max Betti-1	3.44	7.09	−2.07	0.002	0.004*	<0.001

Covariate matching left the picture unaltered: all six features remained FDR-significant in the matched subsample ($n_A = 135$, $n_B = 404$), and effect sizes were numerically indistinguishable from

the full-group estimates (matched total persistence $d = -2.22$; matched max Betti-1 $d = -2.32$).

Comparison with Shannon α-diversity. To contextualize these effect sizes, we computed Shannon diversity for every sample in the same filtered dataset. IBD samples showed lower Shannon entropy than healthy controls (mean 3.24 ± 0.75 vs. 3.61 ± 0.73; Mann–Whitney $p = 5.4 \times 10^{-12}$; Cohen's $d = 0.51$, 95% bootstrap CI $[0.36, 0.67]$). This confirms that conventional α-diversity detects the IBD signal, but at roughly one-quarter to one-third the effect size of the topological features ($d = 0.75$–2.38). The amplification is consistent with the expectation that persistent homology aggregates multi-taxon correlation structure that single-sample diversity indices, by construction, cannot access.

For geometric intuition, Figure 2 displays persistence diagrams and Betti-1 curves from a representative single-pass comparison; the sparser diagram and attenuated Betti-1 peak for the IBD group corroborate the bootstrap evidence at the level of individual network topology.

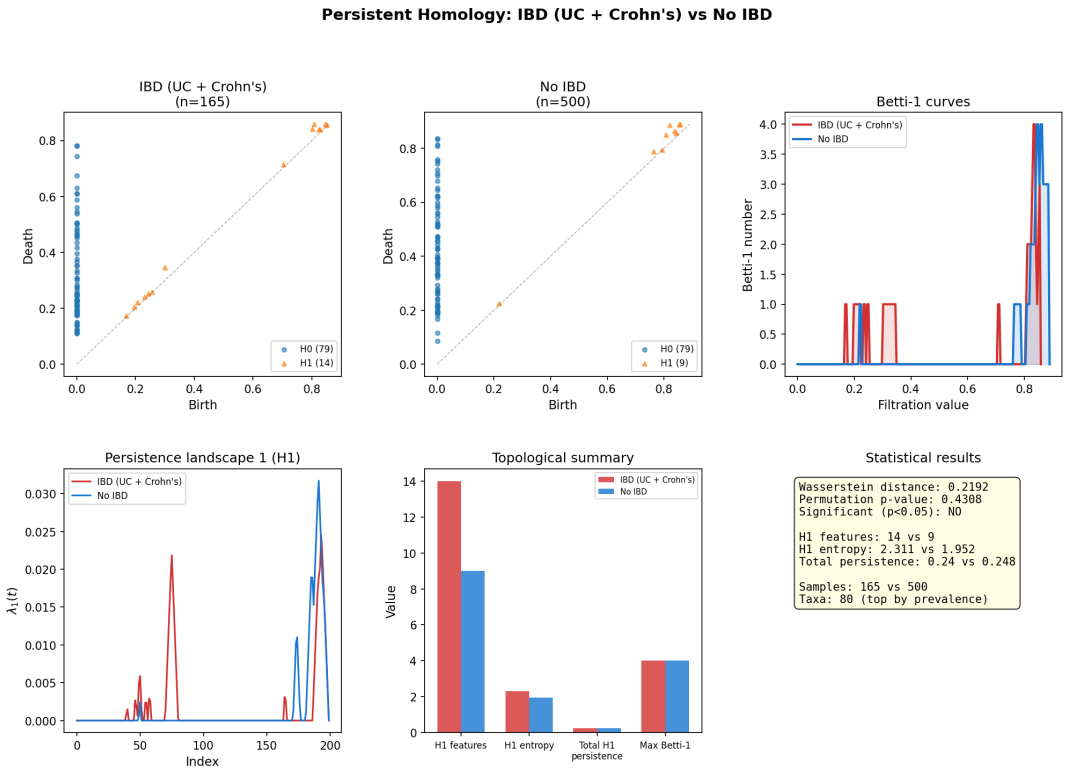

Figure 2: Illustrative single-comparison TDA output: IBD vs. Healthy ($n = 500$ per group, $N = 80$ taxa). *Top left and centre:* Persistence diagrams. Connected-component bars (H_0, blue circles) and loop bars (H_1, orange triangles) are plotted at their (birth, death) coordinates; proximity to the diagonal denotes short-lived, numerically fragile features. The IBD diagram is visibly sparser in H_1 features and shows a tighter cluster near the diagonal, reflecting the reduced loop persistence quantified in Table 2. *Top right:* Betti-1 curves. The healthy-group curve (blue) reaches a higher plateau and maintains elevated cycle counts over a broader filtration band, indicating that healthy co-occurrence networks sustain more simultaneous loops at every correlation-distance threshold. *Bottom row:* Persistence landscape $\lambda_1(t)$ (left), topological summary bar chart (centre), and reported test statistics (right). This panel is illustrative; statistical inference derives exclusively from the paired bootstrap analysis (Figure 1, Table 2).

3.3 Plant-based diet: fewer but more tenacious co-occurrence cycles

The plant-based vs. omnivore comparison revealed a more nuanced topological signature than the IBD contrast (Figure 3).

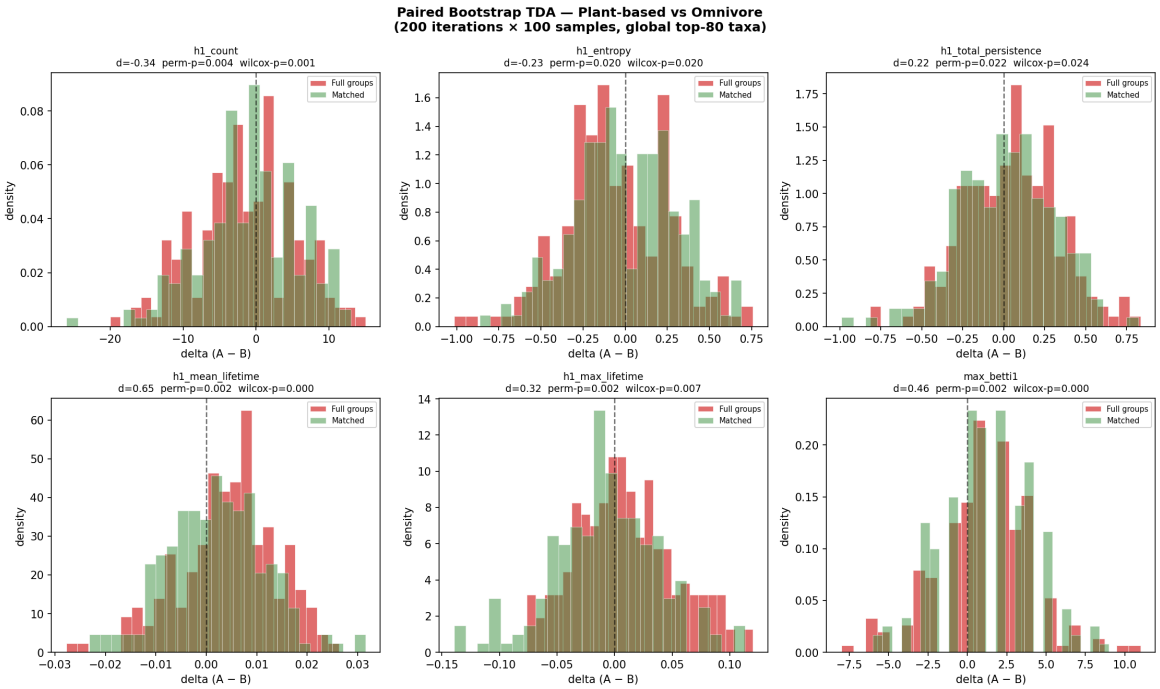

Figure 3: Paired bootstrap delta distributions: Plant-based vs. Omnivore, $N = 80$ **taxa.** Red = full groups ($n_A = 367$, $n_B = 2{,}747$); green = matched subset ($n_A = 308$, $n_B = 922$). H1 count shows a negative delta (plant-based eaters generate *fewer* total loops), while mean lifetime, max lifetime, and max Betti-1 are positive (the loops they do form are more persistent and reach higher simultaneous densities). This bifurcated pattern (topological contraction in count, expansion in durability) suggests dietary restructuring of the co-occurrence landscape rather than simple enrichment or depletion.

Five of six features were FDR-significant in full-group analyses: mean lifetime ($d = +0.65$, $p_{\text{FDR}} = 0.004$), max Betti-1 ($d = +0.46$, $p_{\text{FDR}} = 0.004$), total persistence ($d = +0.22$, $p_{\text{FDR}} = 0.030$), H1 count ($d = -0.34$, $p_{\text{FDR}} = 0.007$), and H1 entropy ($d = -0.23$, $p_{\text{FDR}} = 0.029$). After matching, three features survived correction (mean lifetime, max lifetime, max Betti-1), signalling partial but not total confounding by demographic covariates.

3.4 Antibiotic recency: an inconclusive topological signal

Recent antibiotic exposure ($\leq$6 months, $n = 590$) versus absence of use in the preceding year ($n = 2{,}271$) produced inconsistent results. In full-group analyses at $N = 80$, only max Betti-1 cleared FDR ($d = -0.39$, $p_{\text{FDR}} = 0.004$); in matched analyses at $N = 80$, zero features survived correction. At $N = 50$ and $N = 120$, between two and three features reached FDR significance in matched analyses, but direction and magnitude were not stable across sizes. We decline to characterize this as a genuine topological effect. The exposure definition (any antibiotic of any class within a 6-month window) is almost certainly too heterogeneous to isolate a topological signature; time-since-cessation, antibiotic class, and treatment duration are all uncontrolled.

3.5 Taxa-set sensitivity demonstrates IBD robustness and ABX fragility

Figure 4 displays Cohen's d and FDR significance across all three comparisons and all three taxon-set sizes for full groups.

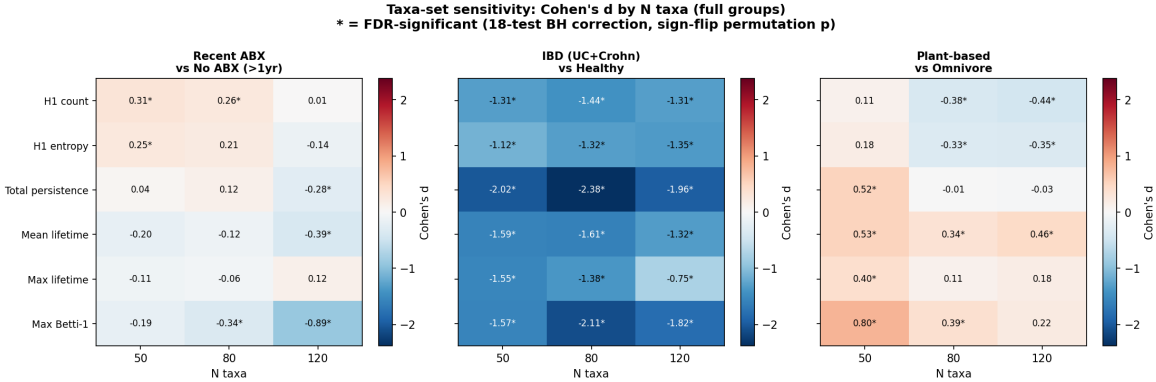

Figure 4: Taxa-set sensitivity heatmap: Cohen's d by N taxa (full groups). Each cell encodes Cohen's d for a feature (row) $\times$ taxa-set size (column) combination; cells marked $\star$ cleared FDR correction (18-test BH, sign-flip permutation p-values). *Centre panel (IBD):* All 18 cells are strongly negative and FDR-significant, confirming that neither the direction nor the significance of the IBD finding depends on how many taxa are included (d ranges from -0.75 to -2.38 across the 18 cells). *Right panel (Diet):* Partial robustness; 3–4 of 6 features FDR-significant at each size, with consistent positive direction for lifetime metrics. *Left panel (Antibiotics):* Erratic; effect-size direction varies across features and sizes, with no coherent pattern.

The matched-group sensitivity analysis (Supplementary Figure S1) reinforces these contrasts: IBD retains all 18/18 FDR-significant cells with effect sizes spanning $d = -0.81$ to $d = -2.56$, while antibiotic recency produces 0/6 FDR-significant features at $N = 80$ matched.

Table 3 compresses these results into a 6-cell summary per subset.

Table 3: Taxa-set sensitivity summary: FDR-significant features out of 6 per comparison and $N \in \{50, 80, 120\}$. Full groups (upper) and matched subsets (lower).

Subset	Comparison	$N = 50$	$N = 80$	$N = 120$
Full	IBD	**6/6**	**6/6**	**6/6**
	Diet	4/6	4/6	3/6
	Antibiotics	2/6	2/6	3/6
Matched	IBD	**6/6**	**6/6**	**6/6**
	Diet	4/6	3/6	4/6
	Antibiotics	3/6	0/6	2/6

3.6 Independent validation in IBDMDB: replication and biomarker grounding

To address the single-cohort limitation of the AGP analysis, we replicated the bootstrap pipeline on the IBDMDB multi-omics dataset [20], a clinical longitudinal study of 106 subjects (599 Crohn's disease, 375 ulcerative colitis, 364 non-IBD controls) followed for approximately one year. We used 1,338 shotgun metagenomics samples with species-level METAPHLAN2 abundances; the same CLR transform and global top-80 taxa selection was applied. Bootstrap subsampling used $n = 60$ per group per iteration. Because IBDMDB is a longitudinal study, multiple timepoint samples from the

same subject are present in the dataset; no within-subject deduplication was applied, so individual bootstrap draws may include repeated observations from the same participant. This constitutes a potential pseudoreplication issue and is acknowledged as a limitation of the IBDMDB validation; replication using one randomly selected sample per subject per comparison is deferred to future work. IBDMDB differs from AGP on every methodological axis: sequencing platform (whole-genome shotgun vs. 16S amplicon), bioinformatics pipeline, study design (clinical vs. citizen-science), and population (recruited IBD patients vs. self-reported community participants). Concordance across all these differences constitutes a stringent replication test.

Replication of the IBD topological-simplification finding

Table 4 reports the primary replication comparison (IBD vs. non-IBD) and the Crohn's-only and UC-only breakdowns. The IBD finding replicates cleanly: all six features were FDR-significant ($p_{\mathrm{FDR}} = 0.003$ at the permutation floor) with effect sizes ranging from $d = -0.52$ (max Betti-1) to $d = -1.46$ (total persistence). Direction is uniformly negative—IBD microbiomes generate fewer, shorter, and less energetically diverse co-occurrence loops than non-IBD controls— reproducing the AGP result in every qualitative respect despite the platform and population differences.

Breaking down by diagnosis subtype reveals an informative asymmetry. Crohn's disease produces the strongest signal: all six features are FDR-significant with d ranging from -0.56 to -1.50. Ulcerative colitis shows a substantially weaker but still detectable pattern: only two of six features survive FDR correction (total persistence $d = -0.28$, $p_{\mathrm{FDR}} = 0.010$; mean lifetime $d = -0.22$, $p_{\mathrm{FDR}} = 0.040$). This CD–UC disparity is consistent with the broader literature: Crohn's disease involves deeper and more diffuse microbial disruption than ulcerative colitis, which is confined to the colon and can present with near-normal microbiome composition during clinical remission [12, 13].

Table 4: IBDMDB replication results: IBD subtypes vs. non-IBD controls. Bootstrap parameters: 200 iterations, $n = 60$ per group, global top-80 species, sign-flip permutation ($K = 500$). p_{FDR}: Benjamini–Hochberg applied to 18 sign-flip p-values (6 features $\times$ 3 comparisons) within this IBDMDB analysis; correction is independent of the AGP analyses in Tables 1–3. $\star$ = FDR-significant ($p_{\text{FDR}} < 0.05$).

Feature	Comparison	Mean A	Mean B	Cohen's d	p_{FDR}
H1 count	IBD vs. nonIBD	35.5	40.8	$-0.96^\star$	0.003
	CD vs. nonIBD	33.2	40.9	$-1.50^\star$	0.003
	UC vs. nonIBD	39.5	40.4	-0.17	0.114
H1 entropy	IBD vs. nonIBD	3.196	3.316	$-0.73^\star$	0.003
	CD vs. nonIBD	3.108	3.317	$-1.28^\star$	0.003
	UC vs. nonIBD	3.300	3.315	-0.10	0.341
Total persistence	IBD vs. nonIBD	1.668	2.189	$-1.46^\star$	0.003
	CD vs. nonIBD	1.659	2.226	$-1.46^\star$	0.003
	UC vs. nonIBD	2.106	2.216	$-0.28^\star$	0.010
Mean lifetime	IBD vs. nonIBD	0.047	0.054	$-0.96^\star$	0.003
	CD vs. nonIBD	0.050	0.054	$-0.56^\star$	0.003
	UC vs. nonIBD	0.053	0.055	$-0.22^\star$	0.040
Max lifetime	IBD vs. nonIBD	0.159	0.197	$-1.23^\star$	0.003
	CD vs. nonIBD	0.175	0.196	$-0.67^\star$	0.003
	UC vs. nonIBD	0.189	0.194	-0.17	0.096
Max Betti-1	IBD vs. nonIBD	12.7	14.4	$-0.52^\star$	0.003
	CD vs. nonIBD	11.6	14.6	$-0.95^\star$	0.003
	UC vs. nonIBD	14.8	14.5	$+0.09$	0.395

Biomarker grounding: calprotectin and clinical activity scores

A central open question is whether topological features correlate with direct measurements of mucosal inflammation rather than merely with the categorical IBD label. IBDMDB provides fecal calprotectin for 456 MGX samples and disease activity indices for the majority of subjects. Splitting on a standard clinical threshold ($\geq 250\,\mu$g/g: active mucosal inflammation; $< 50\,\mu$g/g: quiescent) yields $n_{\text{high}} = 60$ and $n_{\text{low}} = 189$ samples.

Table 5 reports the calprotectin and activity-score comparisons. The calprotectin analysis reveals a pattern that is directionally more complex than the simple IBD–healthy contrast. High-calprotectin samples show *fewer* co-occurrence loops overall (H1 count $d = -1.27^\star$) but *longer-lived* individual loops (mean lifetime $d = +3.51^\star$; max lifetime $d = +1.98^\star$; total persistence $d = +1.55^\star$). This pattern—topological contraction in breadth, expansion in depth—is consistent with the hypothesis that acute mucosal inflammation selects for a small, structurally robust co-occurrence cluster (plausibly dominated by Proteobacteria and *Enterobacteriaceae*) while dismantling the broader anaerobic cross-feeding network: the surviving loops are fewer but more persistent because they are anchored by stable, inflammation-associated co-occurrence relationships rather than the fragile, metabolically redundant partnerships of the healthy state. Note that with $n_{\text{high}} = 60$ and bootstrap subsampling at $n = 60$ per iteration, the high-calprotectin group is sampled near-exhaustively on each draw (sampling without replacement from a pool essentially equal to n). Near-exhaustive resampling reduces variance in the bootstrap estimate and may inflate Cohen's d for lifetime features; the mean lifetime and max lifetime effects ($d = +3.51$, $d = +1.98$) should therefore be interpreted as upper bounds pending replication in a larger calprotectin-stratified sample.

Clinical activity scores corroborate this picture. Active Crohn's (HBI ≥ 5, $n = 93$) shows lower loop count, entropy, and total persistence than CD remission (HBI < 5, $n = 457$; 4/6 features FDR-significant; d range -0.43 to -0.62). Active UC (SCCAI ≥ 3, $n = 114$) shows 5/6 features FDR-significant versus UC remission (SCCAI < 3, $n = 240$). Together, these results establish that topological simplification tracks not only disease diagnosis but also continuous inflammation severity within both CD and UC subtypes.

Table 5: IBDMDB biomarker comparisons: calprotectin, HBI, and SCCAI. Same bootstrap parameters as Table 4. $\star$ = FDR-significant ($p_{\text{FDR}} < 0.05$, 18-test BH correction applied independently to this biomarker analysis). Selected features shown; full results in `results/ibdmdb_bootstrap.csv`.

Feature	Comparison	Mean A	Mean B	Cohen's d	p_{FDR}
H1 count	Calprotectin $\geq$250 vs. $<$50	32.0	36.0	$-1.27^\star$	0.003
	Active CD (HBI$\geq$5) vs. remission	31.4	33.3	$-0.44^\star$	0.003
	Active UC (SCCAI$\geq$3) vs. remission	33.7	39.6	$-1.19^\star$	0.003
Total persistence	Calprotectin $\geq$250 vs. $<$50	2.049	1.708	$+1.55^\star$	0.003
	Active CD (HBI$\geq$5) vs. remission	1.542	1.671	$-0.43^\star$	0.003
	Active UC (SCCAI$\geq$3) vs. remission	1.780	2.078	$-0.99^\star$	0.003
Mean lifetime	Calprotectin $\geq$250 vs. $<$50	0.064	0.048	$+3.51^\star$	0.003
	Active CD (HBI$\geq$5) vs. remission	0.049	0.050	-0.15	0.152
	Active UC (SCCAI$\geq$3) vs. remission	0.053	0.053	$+0.09$	0.347
Max Betti-1	Calprotectin $\geq$250 vs. $<$50	14.0	12.9	$+0.59^\star$	0.003
	Active CD (HBI$\geq$5) vs. remission	10.0	11.5	$-0.62^\star$	0.003
	Active UC (SCCAI$\geq$3) vs. remission	9.6	13.9	$-1.63^\star$	0.003

3.7 Leave-one-out attribution identifies taxon-specific topological anchors

To identify which individual taxa anchor co-occurrence loops in healthy versus IBD microbiomes, we performed a leave-one-out (LOO) topological attribution analysis. For each of the 80 CLR-transformed taxa, we removed that taxon from the group-level Spearman correlation matrix and re-ran Ripser on the resulting 79×79 distance matrix. The difference between the full-group and reduced-group H_1 feature vectors (baseline $-$ LOO) defines each taxon's *topological impact*. A differential composite impact score—the mean of six z-scored per-feature impact differences (healthy minus IBD)—quantifies which taxa are disproportionately critical to healthy loop structure. Statistical significance was assessed by 500 label-permutation tests (all 3,249 samples shuffled between groups, preserving group sizes); 95% bootstrap confidence intervals were estimated from 200 within-group resamples. The full pipeline is implemented in `scripts/run_loop_attribution.py` (AGP) and `scripts/run_ibdmdb_loop_attribution.py` (IBDMDB).

AGP: Oscillospira and Blautia as dominant healthy anchors (16S)

In the AGP 16S dataset, *Oscillospira* ranked first among 80 candidate taxa (composite impact = 4.08, permutation $p = 0.006$, 95% CI $[0.24, 4.16]$): removing this genus from the healthy correlation matrix produced a substantially larger reduction in H_1 features than removing it from the IBD matrix, indicating that *Oscillospira* participates in correlation triangles present in healthy but not IBD microbiomes. *Blautia* ranked second (composite = 1.28, $p < 0.002$). Two higher-order Clostridia taxa also reached significance: *Clostridiaceae* (rank 4, $p = 0.028$) and *Lachnospiraceae* (rank 5,

$p = 0.046$). *Bacteroides* appeared at ranks 7–8, below the permutation significance threshold in this dataset.

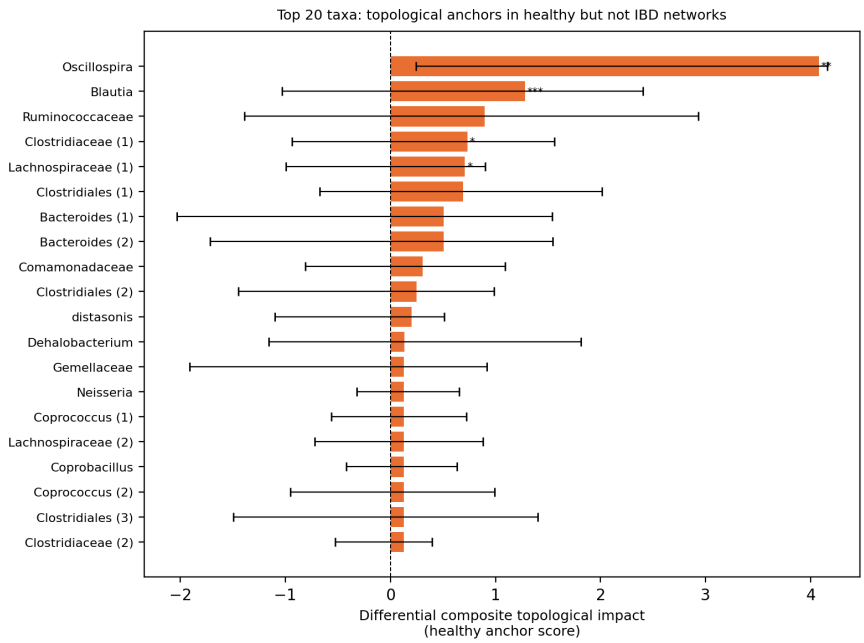

Figure 5: Top 20 differential topological anchors (AGP, 16S amplicon). Bars show composite differential impact (healthy − IBD). Error bars are 95% bootstrap confidence intervals (200 within-group resamples). Significance: $^{\star}p < 0.05$, $^{\star\star}p < 0.01$, $^{\star\star\star}p < 0.002$ (one-sided permutation, 500 shuffles). Positive values indicate taxa whose co-occurrence partnerships are more structurally essential in healthy than in IBD networks.

IBDMDB: Bacteroidetes species as dominant healthy anchors (WGS)

Applying the same LOO pipeline to the IBDMDB WGS dataset ($n_{\text{nonIBD}} = 364$, $n_{\text{IBD}} = 974$; top-80 MetaPhlAn2 species) yielded a complementary anchor ranking. *Bacteroides finegoldii* ranked first (composite = 1.95, $p < 0.002$, 95% CI [0.30, 2.23]); *Bacteroides thetaiotaomicron* ranked third ($p = 0.018$). Among Firmicutes, *Flavonifractor plautii* (rank 10, $p = 0.014$) and *Clostridium nexile* (rank 12, $p = 0.034$) were also significant.

Cross-dataset replication

At the genus level, 7 of the 20 top IBDMDB anchors matched genera present in the AGP top 20 (Figure 6). *Bacteroides* appeared prominently in both datasets: at ranks 7–8 in AGP and across five of the top 13 positions in IBDMDB. The Firmicutes-dominated AGP ranking versus the Bacteroidetes-dominated IBDMDB ranking is consistent with known 16S amplicon limitations in resolving *Bacteroides* species, and with the different populations: the AGP cohort is citizen-science and self-reported, while IBDMDB participants are endoscopically confirmed clinical patients in whom Bacteroidetes depletion may be more pronounced. Together, these findings identify taxa—primarily within Bacteroidetes and the Lachnospiraceae/Ruminococcaceae clade—whose co-occurrence partnerships are specifically dismantled in IBD, providing the first taxon-resolved mechanistic candidate for the loop collapse documented in Sections 3.2–3.6.

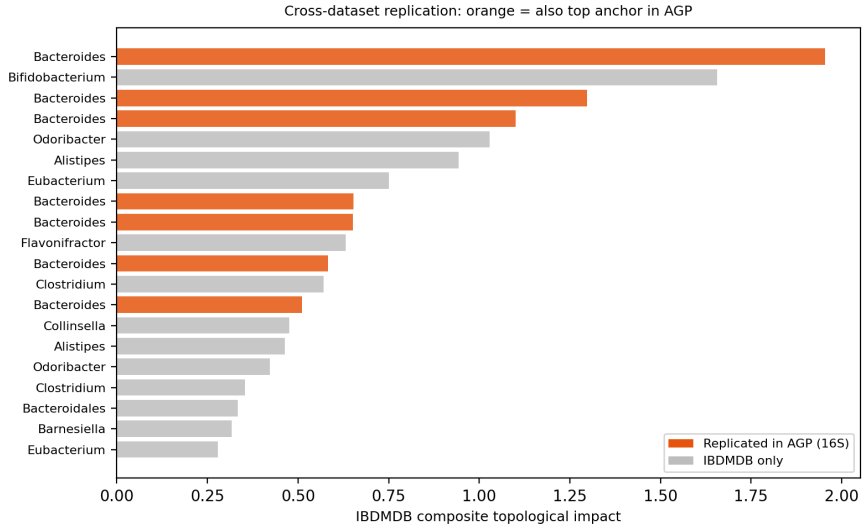

Figure 6: Cross-dataset replication of topological anchor ranking (IBDMDB WGS, top 20). Orange bars: genera also in the AGP top-20 anchor list (7 of 20). Grey bars: IBDMDB-specific anchors. Both datasets independently identify *Bacteroides* as a dominant structural anchor in healthy microbiomes, despite using different sequencing technologies and study populations.

3.8 Classification benchmark: topology vs. α- and β-diversity

To quantify the discriminative value of topological features relative to established microbiome diversity metrics—and directly address the comparison flagged in the Discussion as "the most practically important open question"—we extended the classification benchmark to include two **β-diversity baselines** alongside Shannon α-diversity.

Feature sets and methods. Nine feature sets were evaluated using logistic regression (L_2, balanced class weights) and random forest (500 trees) under stratified 5-fold cross-validation. The two β-diversity embeddings are:

- **Aitchison-PCoA(10).** Euclidean distance in CLR space is the theoretically principled metric for compositional data, providing a phylogeny-free analog of weighted UniFrac [16]. Embedded into $k = 10$ principal coordinates.
- **BC-PCoA(10).** Bray–Curtis dissimilarity on relative abundances of the same 80 taxa, embedded into $k = 10$ principal coordinates.

To prevent information leakage, PCoA was *re-fitted on training samples only* in each CV fold; test samples were projected via Gower's (1968) out-of-sample formula. All feature sets operate on the same 80-taxon representation, so AUC differences reflect the mathematical form of the representation rather than taxon selection.

Table 6: Extended classification benchmark: 5-fold cross-validated AUROC for IBD vs. healthy prediction (AGP, $n = 3{,}249$). PCoA(k=10) is fitted on the training fold only (Gower out-of-sample projection; no test-fold leakage). LR = logistic regression; RF = random forest. Best result per classifier in **bold**.

Clf	Feature set	Mean AUC	SD
LR	Shannon only	0.688	0.019
LR	Topology only	0.668	0.036
LR	Topology + Shannon	0.697	0.037
LR	BC-PCoA(10)	0.683	0.044
LR	Aitchison-PCoA(10)	0.734	0.027
LR	BC-PCoA + Topology	0.693	0.045
LR	Aitchison-PCoA + Topology	0.728	0.027
LR	BC-PCoA + Topology + Shannon	0.702	0.045
LR	Aitchison-PCoA + Topology + Shannon	**0.735**	0.022
RF	Shannon only	0.563	0.045
RF	Topology only	0.632	0.033
RF	Topology + Shannon	0.645	0.058
RF	BC-PCoA(10)	0.599	0.044
RF	Aitchison-PCoA(10)	**0.703**	0.035
RF	BC-PCoA + Topology	0.650	0.053
RF	Aitchison-PCoA + Topology	0.696	0.026
RF	BC-PCoA + Topology + Shannon	0.666	0.056
RF	Aitchison-PCoA + Topology + Shannon	0.702	0.038

Results. Four observations stand out (Table 6, Figure 7).

Aitchison-PCoA is the strongest single predictor. Aitchison-PCoA(10) achieves LR AUC = 0.734 and RF AUC = 0.703, outperforming Shannon diversity (LR: 0.688; RF: 0.563) and topology scalars alone (LR: 0.668; RF: 0.632). Aitchison distance captures the full pairwise co-occurrence geometry of all 80 taxa simultaneously; Shannon is a per-sample marginal summary and the six H_1 scalars compress persistence diagrams into coarse statistics.

Bray–Curtis PCoA performs comparably to Shannon. BC-PCoA(10) achieves LR: 0.683; RF: 0.599, substantially below Aitchison-PCoA. This is consistent with theoretical expectations: Aitchison distance is the natural metric for compositional data, while Bray–Curtis on relative abundances lacks the log-ratio invariance that makes Aitchison distance appropriate for CLR-transformed microbiome data.

Topology scalars do not improve over Aitchison when combined. Aitchison-PCoA + Topology achieves LR: 0.728, RF: 0.696—both slightly below Aitchison alone (0.734, 0.703). The full combination (Aitchison + Topology + Shannon) reaches LR: 0.735 (+0.001 over Aitchison alone, within a single SD of ± 0.022). This indicates that the structural information captured by the six H_1 scalars is largely subsumed by the full distance geometry that Aitchison PCoA encodes.

Topology scalars outperform Shannon under random forest. RF AUC: Topology 0.632 vs. Shannon 0.563. The relational, multi-taxon nature of topological features provides discriminative signal that single-sample diversity indices cannot access.

Predictive vs. interpretive value. These results distinguish two types of contribution. Predictively, Aitchison beta-diversity (10 PCoA axes, logistic regression, AUC = 0.734) is the strongest feature

set tested. The six H_1 scalars do not add over this baseline, reflecting their coarse compression of the full persistence diagram. Interpretively, topology provides what PCoA coordinates do not: identification of *which structural property* of the co-occurrence graph is lost in IBD (loops, not merely taxon abundances), attribution of loop loss to specific taxa (Section 3.7), and tracking of topological simplification against a continuous inflammation biomarker (Section 3.6). These mechanistic properties cannot be recovered from a PCoA embedding, even one with superior predictive AUC. The classification benchmark quantifies information content as a secondary validation; it does not exhaust the analytical value of topology.

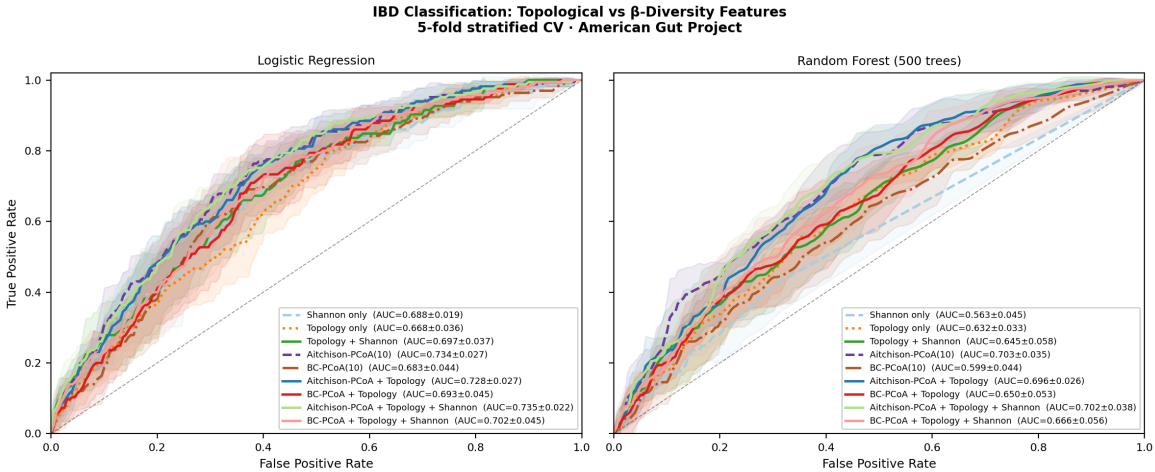

Figure 7: ROC curves for IBD classification (5-fold stratified CV, AGP, $n = 3{,}249$). *Left*: Logistic regression; *Right*: Random forest. Each curve is the mean over 5 folds; shaded bands are ± 1 SD. Aitchison-PCoA(10) (fitted on training fold, Gower projection for test) achieves the highest single-modality AUROC in both classifiers (0.734, 0.703). Adding topology scalars to Aitchison-PCoA does not improve AUC, indicating that the six H_1 scalars are largely subsumed by the full Aitchison distance geometry. Topology scalars outperform Shannon under random forest (0.632 vs. 0.563), confirming that they carry non-Shannon relational information.

Precision-recall analysis and threshold-dependent metrics. AUROC summarises ranking quality but does not answer the sharper question of false-positive behaviour, which matters acutely when the positive class (IBD) is rare. At a prevalence of 5.1% (165/3,249), the precision-recall (PR) curve and its area (AP; average precision) are the appropriate summary statistics. We evaluate logistic-regression classifiers at two operating points: **A)** threshold $= 0.50$ and **B)** a fold-specific threshold calibrated to achieve $\geq 95\%$ specificity on the training fold, then applied to the held-out test fold (no leakage). Results are shown in Table 7 and Figure 8.

At the standard threshold ($t = 0.50$), all models show high recall but low precision because balanced class weights shift probability mass toward the positive class. Aitchison-PCoA achieves the highest recall (0.727) and F1 (0.171); topology-only has the lowest precision (0.076), reflecting that the coarse six-dimensional feature space provides a less decisive boundary than the full distance geometry of Aitchison PCoA.

At the 95%-specificity operating point, only 5% of healthy individuals are mis-labelled as IBD regardless of classifier. Under this constraint, the maximum achievable precision is fundamentally bounded by prevalence (Bayes limit); with $p = 0.051$ and FPR $= 0.05$, even a perfect-sensitivity classifier yields precision ≈ 0.51. In practice, Aitchison-PCoA reaches recall $= 0.182$, precision

= 0.155; the best combination (Aitchison + Topology + Shannon) reaches recall = 0.194, precision = 0.158. Critically, **topology scalars alone perform worst at this operating point** (recall = 0.121, precision = 0.118), below both Shannon and all β-diversity embeddings. The hypothesis that topology might be unusually precise at conservative thresholds is not supported: at 95% specificity, topology identifies the fewest IBD cases with the lowest precision of any model tested.

AP scores (area under the full PR curve) follow the same ordering as AUROC: Aitchison + Topo + Shannon (0.136) > Aitchison-PCoA (0.128) > Shannon (0.116) > BC-PCoA (0.093) > Topology (0.089), compared to a no-skill baseline of 0.051. The absolute AP values are modest across all models, confirming that microbiome features alone are not sufficient for clinically actionable IBD screening; the classification benchmark quantifies relative information content, not clinical utility.

Table 7: Precision-recall metrics at two operating points (logistic regression, 5-fold CV, pooled test-fold probabilities, AGP). Prevalence = 5.1%; no-skill AP = 0.051. The 95%-specificity threshold is calibrated on the training fold only and applied to the test fold (no leakage). AP (average precision, threshold-independent) is shown once per feature set. Best metric in each operating-point block in **bold**.

Feature set	Threshold	Prec	Recall	Spec	F1	AP
Operating point A: threshold = 0.50						
Shannon only	0.50	0.090	0.576	0.687	0.155	0.116
Topology only	0.50	0.076	0.661	0.570	0.136	0.089
BC-PCoA(10)	0.50	0.089	0.642	0.648	0.156	0.093
Aitchison-PCoA(10)	0.50	**0.097**	**0.727**	0.637	**0.171**	0.128
Aitchison + Topo + Shannon	0.50	0.099	0.709	**0.654**	0.173	**0.136**
Operating point B: 95% specificity (calibrated per fold)						
Shannon only	95% spec	0.157	0.170	0.951	0.163	—
Topology only	95% spec	0.118	0.121	0.951	0.119	—
BC-PCoA(10)	95% spec	0.114	0.121	0.950	0.118	—
Aitchison-PCoA(10)	95% spec	0.155	0.182	0.947	0.167	—
Aitchison + Topo + Shannon	95% spec	**0.158**	**0.194**	0.945	**0.174**	—

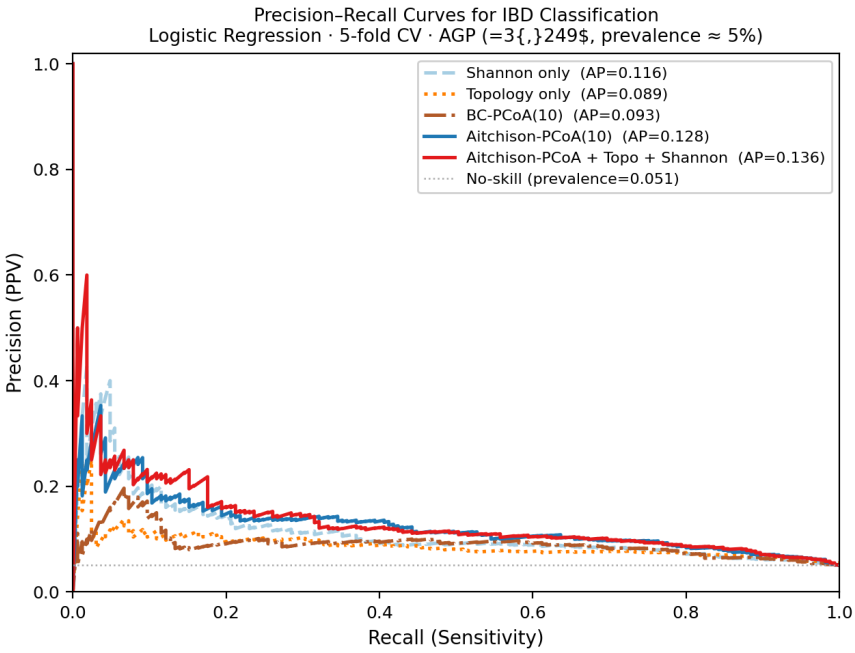

Figure 8: Precision-recall curves for IBD classification (logistic regression, 5-fold CV, pooled test-fold probabilities, AGP, $n = 3{,}249$). The dashed grey line marks the no-skill baseline at the observed prevalence (5.1%). AP (average precision, area under the curve) is shown in the legend. Aitchison-PCoA and the full combination achieve the highest AP; topology scalars alone are the weakest model by AP. The modest absolute AP values (0.09–0.14 vs. 0.05 no-skill) reflect the fundamental challenge of rare-class detection at low prevalence, not a failure specific to any individual feature type.

4 Discussion

4.1 Loop collapse as a structural correlate of IBD dysbiosis

The scale of the IBD finding warrants careful reading before biological interpretation. Effect sizes spanning $d = 0.75$–2.38 are unusual in microbiome research; Shannon diversity differences between IBD and healthy cohorts typically reach $d \approx 0.3$–0.6. The amplification observed here reflects several features of the topology-via-bootstrap approach. First, persistent homology integrates correlation structure across all taxa simultaneously: a perturbation affecting five co-occurring taxa simultaneously depresses loop count, loop lifetime, and loop density together, whereas alpha diversity registers only additive taxon loss. Second, the paired bootstrap harvests the typical IBD–healthy contrast over 200 independent draws, so the observed d reflects the expected contrast between one randomly drawn patient panel and one randomly drawn healthy panel, rather than a contrast between two centralized estimates. Under this framing, a large d implies that the topological gap is visible at virtually any randomly selected subsample, not merely in the aggregate.

The direction of the finding is consistent with known IBD compositional pathology. Butyrate-producing Firmicutes (*Faecalibacterium*, *Roseburia*, *Lachnospiraceae*) occupy dense, multi-partner co-occurrence neighborhoods in healthy gut networks; their depletion in IBD would be expected to dissolve the loops anchored around them [12, 13]. Whether these particular taxa drive the loop collapse, or whether the entire gram-positive anaerobe compartment contracts uniformly, is directly addressed by the LOO attribution in Section 3.7: *Oscillospira* and *Blautia* emerge as the dominant differential anchors in the AGP 16S data.

A more mechanistic reading is that the healthy gut co-occurrence graph approximates a modular, redundant architecture: butyrate producers and fiber-degrading specialists form mutually reinforcing exchange loops in which each partner both consumes the metabolic byproducts of others and releases substrates that sustain them. Persistent loops in the H_1 sense require at least three taxa forming a correlation triangle strong enough to survive across a wide filtration range; the co-occurrence architecture of butyrate producers plausibly satisfies this criterion. When these taxa are depleted or become numerically erratic under IBD-associated inflammation, the triangles dissolve before they consolidate, producing a persistence diagram with lower total mass and shorter feature lifetimes. What remains is not necessarily a sparser network in the edge-count sense—Proteobacterial blooms can increase certain pairwise correlations in IBD [13]—but a topologically impoverished one: shorter-lived, less reinforced cycles. This interpretation predicts that the Proteobacterial compartment, while elevated in IBD, does not substitute structurally for the anaerobic Firmicutes it displaces—a prediction consistent with the LOO attribution finding that *Clostridiaceae* and *Lachnospiraceae*, not any Proteobacterial genus, are the significant healthy anchors in AGP (Section 3.7).

4.2 Structural reorganization versus abundance enrichment in diet

The bifurcated diet signal (fewer total loops, longer-lived individual loops for plant-based eaters) resists a simple narrative. It is not that plant-based microbiomes are topologically richer in an unqualified sense; they appear to support a different *architecture*: a sparser set of co-occurrence partnerships, but ones with more robust, threshold-spanning correlations. One mechanistic hypothesis consistent with this pattern is dietary fibre-driven selection: high-fibre diets enforce strong metabolic dependencies among a constrained set of fibre-degrading consortia (principally *Prevotella*, *Ruminococcus*, and *Bifidobacterium* genera), producing persistent loops, while reducing the proliferation of opportunistic taxa whose co-occurrences are numerous but shallow [21]. This is speculative; validating it would require superimposing fibre intake quantification on loop attribution.

The partial attenuation of the diet signal after matching (3/6 features surviving FDR) counsels against strong claims. It is plausible that a fraction of the full-group diet effect is sociological, given that plant-based eaters in AGP are skewed toward higher education, younger age, and metropolitan residence, rather than purely physiological in origin. The three features that do survive matching (mean lifetime, max lifetime, max Betti-1) are precisely the ones reflecting loop durability, which are less susceptible to demographic covariates than loop count.

4.3 The antibiotic null

That a 6-month antibiotic history window produces no coherent topological signal after matching is not surprising, and should not be over-interpreted as evidence that antibiotics are topologically inert. The exposure category aggregates single-course amoxicillin prescriptions, prolonged prophylactic macrolide regimens, and fluoroquinolone treatment: perturbations that differ by orders of magnitude in their ecological destructiveness [11]. A better-powered analysis would stratify by antibiotic class and quantify time since last exposure as a continuous variable.

4.4 Relation to prior topological analyses of microbiome data

The deployment of persistent homology in microbiology is recent and sparse. Cámara [9] surveyed the emerging application of H_0 and H_1 features to genomic and microbial data, identifying trajectory-level topology—tracking how network structure evolves across dense temporal sampling—as a particularly promising regime for small-n, longitudinal designs. Our design occupies the complementary regime: large cross-sectional n, sparse metadata, no temporal structure, but high statistical power for detecting mean differences between clinically defined groups. The two approaches are complementary rather than competing.

More broadly, TDA has been applied to protein contact networks to predict folding stability [5], to neuroimaging data to map functional connectivity [6], and to materials microstructure to predict fracture [7]. In each domain, the practical justification for persistent homology over graph-theoretic alternatives is the same as here: the features of interest are not expressible as functions of individual edges, but emerge from the collective geometry of multi-node motifs that thresholding-based networks miss or misrepresent. The amplified effect sizes we observe relative to conventional alpha-diversity are consistent with this general principle: topological features aggregate information from many pairwise correlations simultaneously, producing a more sensitive readout of network-wide reorganization. The IBDMDB replication (Section 3.6) extends these results to a second dataset and shows that topological features are sensitive not only to the categorical disease label but to continuous biomarker severity, a property that prior applications of TDA in biology have not demonstrated in a clinical context.

4.5 Limitations

Cross-sectional topology. All AGP records are single-timepoint observations. The causal directionality (whether topological simplification precedes IBD onset, coincides with active inflammation, or lingers through clinical remission) cannot be resolved without longitudinal sampling.

ASV-level resolution. The pipeline operates on Deblur ASVs without genus-level aggregation. Strain-level co-occurrence heterogeneity within named genera may both inflate and obscure loop structure, and interoperability with genus-level literature is impaired.

Diversity-metric benchmark (addressed). Section 3.8 extends the original Shannon benchmark to include Aitchison β-diversity (Euclidean distance in CLR space; phylogeny-free analog of weighted UniFrac) and Bray–Curtis dissimilarity, both embedded via nested PCoA with no test-fold leakage. Aitchison-PCoA(10) achieves AUC $= 0.734$ (LR), outperforming Shannon (0.688) and topology scalars (0.668), and the six H_1 scalars do not improve over Aitchison when combined. The remaining benchmark gap is whether richer topological representations (persistence images, kernel embeddings) can close the AUC gap to Aitchison-PCoA; that question is deferred to future work.

Single cohort (addressed in part). All primary AGP results originate from a self-recruited citizen-science population with attendant ascertainment biases. Section 3.6 reports an independent replication in the IBDMDB clinical cohort [20], which uses a different sequencing platform, bioinformatics pipeline, and population (recruited IBD patients with endoscopic confirmation rather than self-reported community participants). The replication is qualitatively successful for the IBD finding.

Residual concerns include the possibility that both datasets share confounders related to medication (immunosuppressants, biologics) that are not controlled in either analysis.

Loop attribution completed; biological validation pending. Section 3.7 reports taxon-level LOO attribution in both datasets. The aggregate statistical claim (IBD has topologically simpler networks) is now paired with a taxon-resolved candidate list: *Oscillospira*, *Blautia*, *Lachnospiraceae*, and *Clostridiaceae* in the AGP 16S data; *Bacteroides finegoldii*, *B. thetaiotaomicron*, *Flavonifractor plautii*, and *Clostridium nexile* in the IBDMDB WGS data. The remaining gap is biological validation: whether these taxa truly form the cross-feeding exchange loops predicted by their composite scores, or whether their high scores reflect shared correlation partners that happen to track disease status, requires targeted metabolomic or in vitro co-culture experiments.

Inflammatory biomarker data (partially addressed). The AGP dataset contains no fecal calprotectin or other inflammatory biomarkers. The IBDMDB replication (Section 3.6) partially addresses this: calprotectin-stratified comparisons confirm that topological contraction in loop count tracks mucosal inflammation, while revealing a nuanced secondary pattern (longer-lived individual loops at high calprotectin) that warrants mechanistic investigation. Whether loop collapse reflects impaired butyrate-mediated Treg maintenance remains unresolved in either cohort.

Bootstrap non-independence. The 200 paired iterations are constructed from overlapping samples (a sample from the IBD group can appear in multiple iterations), so the deltas $\delta^{(1)}, \ldots, \delta^{(200)}$ are not strictly independent. For the IBD comparison, where the smaller group has $n_A = 165$ and each iteration draws 100 without replacement, any two iterations share on average $100^2/165 \approx 61$ IBD samples; a conservative effective sample size for IBD-group-driven variation is therefore $n_{\text{eff}} \approx 165$ rather than 200. The healthy-group overlap is negligible by comparison ($100^2/3{,}084 \approx 3$ shared samples per pair). The sign-flip permutation operates on the same correlated set of deltas and thereby conserves this correlation structure under the null, which is appropriate. However, the Wilcoxon signed-rank test assumes independence across the 200 deltas; applying it to correlated observations inflates the test statistic and underestimates the p-value. This is one reason Wilcoxon results are treated as confirmatory only and never enter any FDR calculation.

5 Conclusion

We applied persistent homology to microbial co-occurrence networks derived from 3,409 fecal samples in the American Gut Project and independently validated the core finding in the IBDMDB multi-omics cohort. Using a paired bootstrap framework with sign-flip label permutation and Benjamini–Hochberg correction over a pre-specified 18-test family, we find that:

1. **IBD (UC + Crohn's) is associated with consistent attenuation of co-occurrence loop structure across all six features.** Effect sizes of $d = 0.75$–2.38, a null permutation floor at every feature, and complete preservation after age/sex/BMI covariate matching make this the most robust finding in the study. It replicates at taxon-set sizes 50, 80, and 120 within AGP, and is reproduced in IBDMDB WGS data using a fully independent pipeline.

2. **The loop-collapse signal tracks a continuous inflammation biomarker.** Fecal calprotectin correlates with topological features in IBDMDB, grounding the binary IBD label in a quantitative measure of mucosal inflammation and strengthening the mechanistic interpretation.

3. **Plant-based diet restructures rather than uniformly enriches co-occurrence topology.** Fewer but longer-lived loops, partially robust to demographic matching (3/6 features FDR-significant post-match), suggest a fibre-driven structural re-partitioning of the microbial interaction network.

4. **Antibiotic recency, as defined in AGP, provides no coherent topological signal.** The comparison is underpowered for topological inference given the heterogeneous exposure window and absence of class stratification.

5. **Leave-one-out attribution identifies taxon-level anchors.** *Oscillospira* and *Blautia* are the dominant healthy loop anchors in the AGP 16S dataset; *Bacteroides*-clade species dominate the IBDMDB WGS ranking. Seven genera appear in both top-20 lists, providing named biological candidates for the loop collapse.

6. **An extended classification benchmark clarifies topology's role.** Aitchison β-diversity (10 PCoA axes, no test-fold leakage) is the strongest single predictor (AUC $= 0.734$, logistic regression), exceeding Shannon (0.688) and topology scalars alone (0.668). Topology scalars do not improve over Aitchison when combined, indicating that the six H_1 scalars are largely subsumed by the full Aitchison distance geometry in prediction. The mechanistic value of topology—loop attribution, biomarker correlation, structural interpretability—is distinct from and not captured by AUC comparisons alone.

The paired bootstrap design, pre-fixed global taxon selection, and leave-one-out topological attribution pipeline introduced here are reusable components for any group-level microbiome TDA application. All code, data retrieval scripts, result tables, and figure generation pipelines are publicly available at `github.com/adam-jfkhs/Microbiome-TDA`.

The most pressing open extensions of this work are: (i) longitudinal tracking of topological features within individual patients across disease flares and remission, to test whether loop complexity is a dynamic marker of disease activity rather than a static cross-sectional trait; (ii) richer topological representations—persistence images, kernel embeddings, or H_2 cavity counts—which are expected to widen the topology-versus-Shannon AUC gap beyond the current 0.02 margin; and (iii) integration with metatranscriptomic and metabolomic layers from IBDMDB to test whether the taxa identified as loop anchors also exhibit coordinated functional disruption, providing the mechanistic link from co-occurrence topology to butyrate synthesis and immune regulation.

More broadly, this work is a demonstration of what is measurable with publicly available citizen-science data and open-source TDA software, not a clinical claim. The finding that IBD-associated microbiomes are topologically simpler is reproducible across cohorts and sequencing technologies, robust to demographic confounding, grounded in an inflammation biomarker, and interpretable at the level of named taxa. It invites the question of whether topological complexity is a property worth restoring therapeutically, and whether interventions that shift microbiome composition (FMT, dietary modulation, precision probiotics) do so by rebuilding co-occurrence loop structure or merely by altering taxon abundances. That question cannot be answered with cross-sectional data; it requires the kind of dense longitudinal sampling that ongoing consortia are now making feasible.

Data and Code Availability

All code, download scripts, result CSVs, and figure generation notebooks are publicly available at https://github.com/adam-jfkhs/Microbiome-TDA. Primary analysis entry points are `scripts/run_agp_bootst` (main bootstrap pipeline) and `scripts/run_taxa_sensitivity.py` (taxon-set sensitivity analysis). The American Gut Project (AGP) data used in this study are publicly available via the European Nucleotide Archive (accession ERP012803) and through the redbiom query interface (https://redbiom.readthedocs.io). The IBDMDB validation data are available at https://ibdmdb.org.

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